

ENV 4120

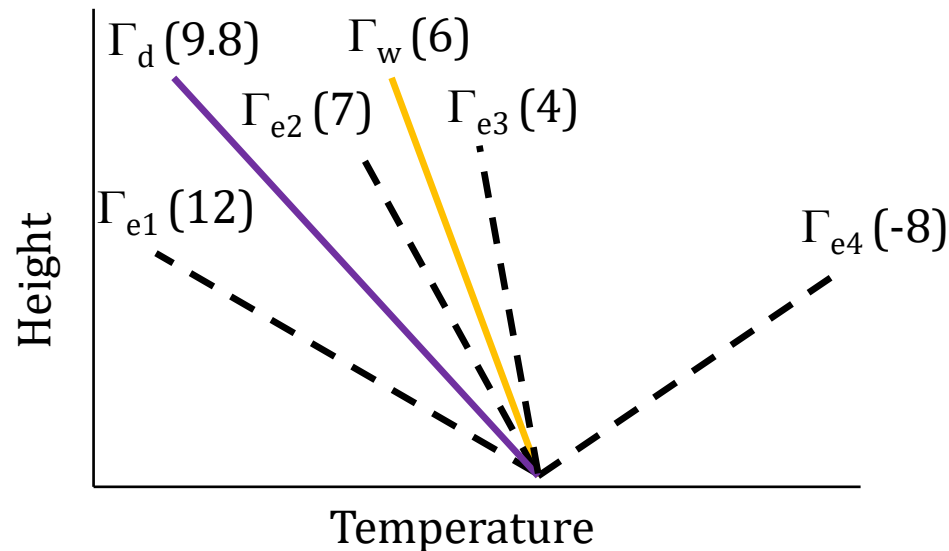
Air Pollution and Hazardous Waste Control

Today's Topics

- Gaussian plume model
- Expectation
 - Understand the concept behind Gaussian plume model and be able to use Gaussian plume model to calculate downwind pollutant concentrations

Quick refresh of stability class

- ☐ $\Gamma_e > \Gamma_d$: Absolutely unstable (Γ_{e1})
- ☐ $\Gamma_e = \Gamma_d$: Dry neutral
- ☐ $\Gamma_d > \Gamma_e > \Gamma_w$: Conditionally unstable (Γ_{e2})
- ☐ $\Gamma_e = \Gamma_w$: Wet neutral
- ☐ $\Gamma_e < \Gamma_w$: Absolutely stable (Γ_{e3}, Γ_{e4})
- Real parcel lapse rate (Γ_p) generally between Γ_d and Γ_w



Categorizing Stability: Classes

day: more stable as u increases
night: less stable as u increases

more stable with less sunlight

more stable with less clouds

Surface Wind				Night-Time Cloudiness	
(measured at 10 m)	Day-Time Insolation			Thinly Overcast or $\geq$ 4/8 Cloudiness	
m s^{-1}	Strong	Moderate	Slight		$\leq 3/8$ Cloudiness
< 2	A	A-B	B	-	-
2-3	A-B	B	C	E	F
3-5	B	B-C	C	D	E
5-6	C	C-D	D	D	D
> 6	C	D	D	D	D

A-extremely unstable, B-moderately unstable, C-slightly unstable, D-neutral, E-slightly stable, F-moderately stable

(Hanna et al., 1982, Handbook on Atmospheric Diffusion. U.S. Dept. of Energy).

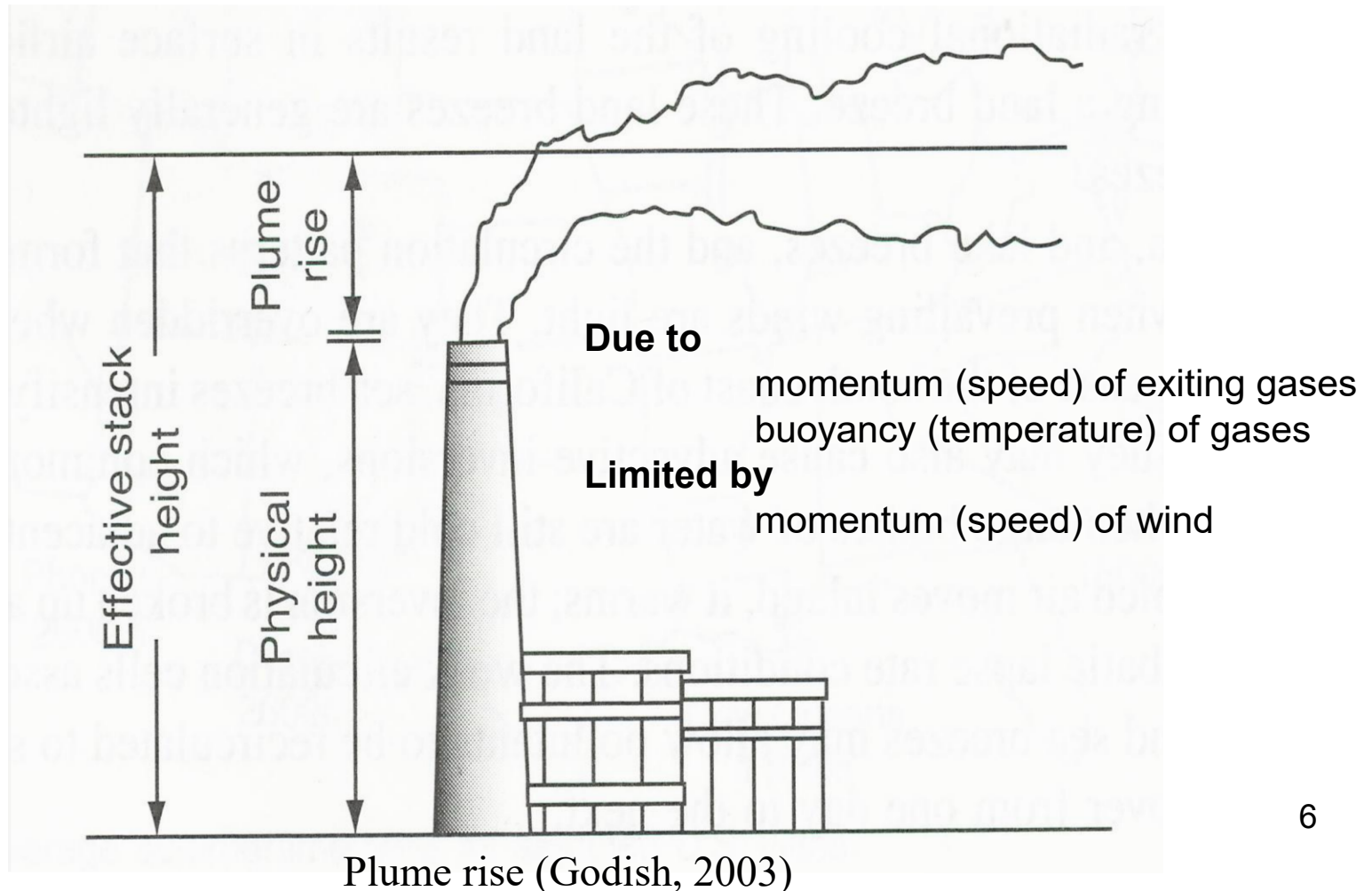
Motivation

- How to calculate pollutant concentrations at one receptor location due to emissions?
- Start with a simple stack....
- First understand the plume from the stack...

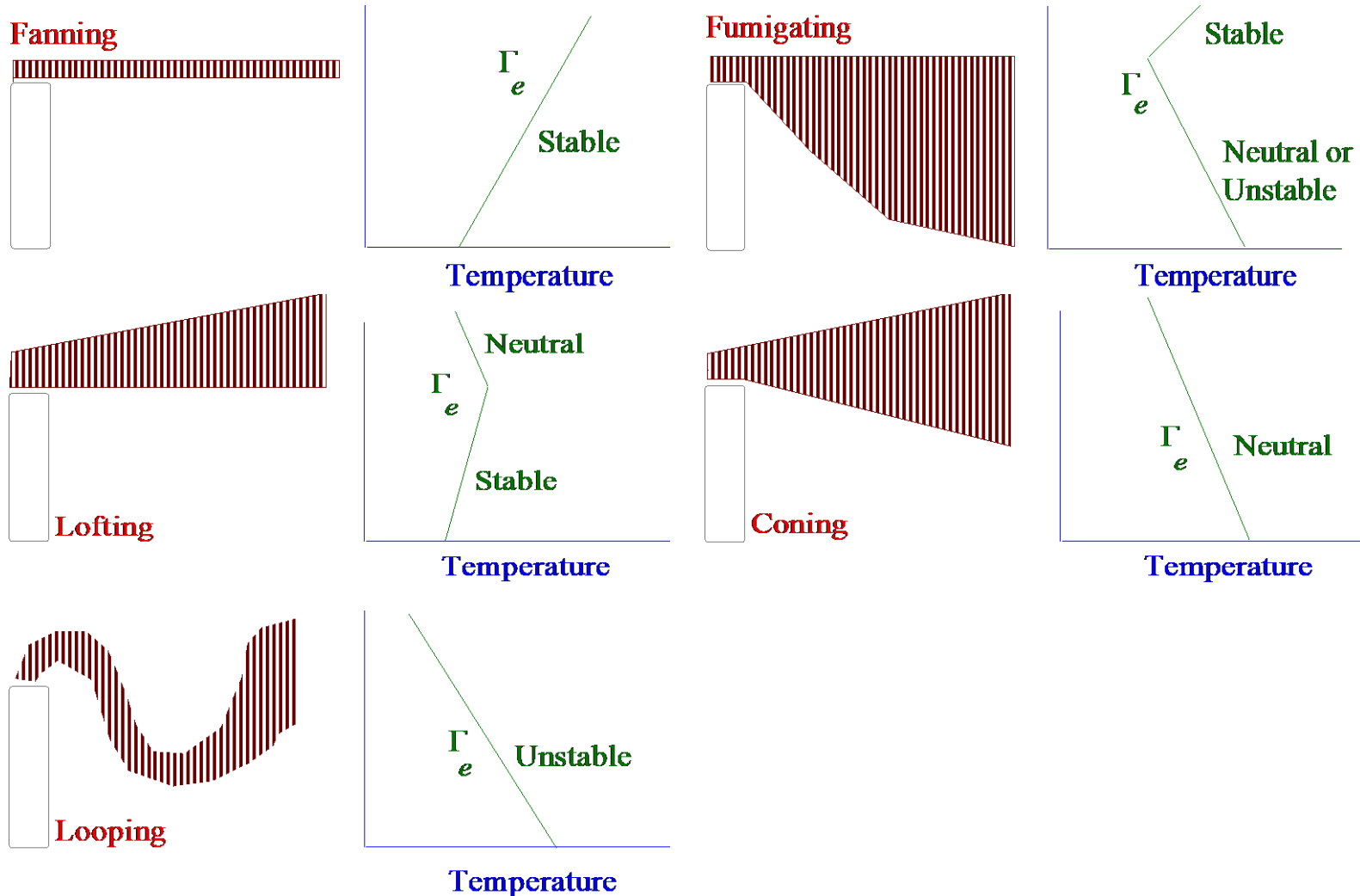


Plume rise

- Plume first rises after release

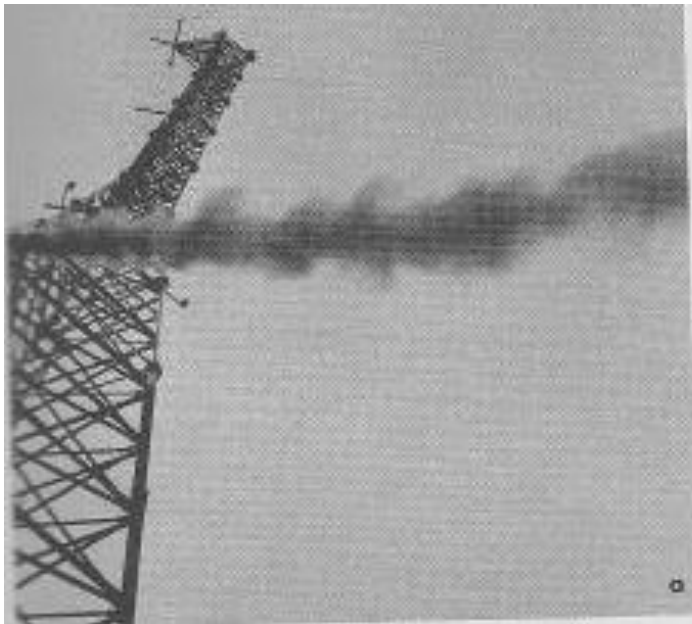


Plume shape vs stability



Time averaged plume

- Instantaneously, plume shape varies
- On average: “Coning” plume



1/25 seconds exposure

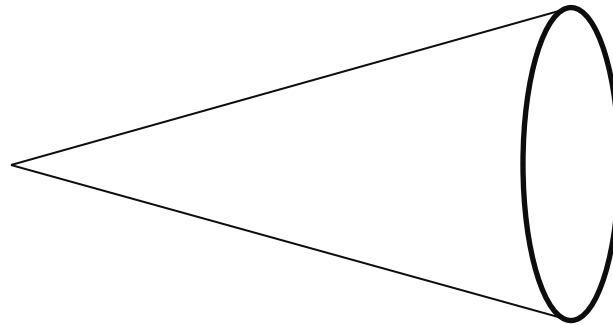


5 minutes exposure

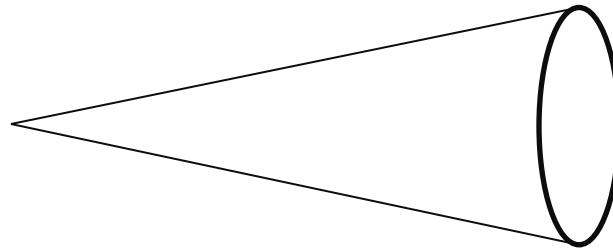
Time averaged plume

- More unstable atmosphere → more spread out

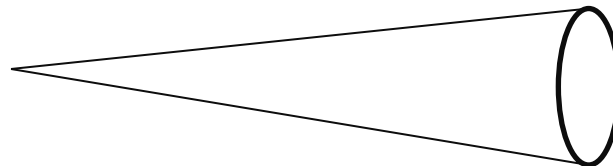
Unstable



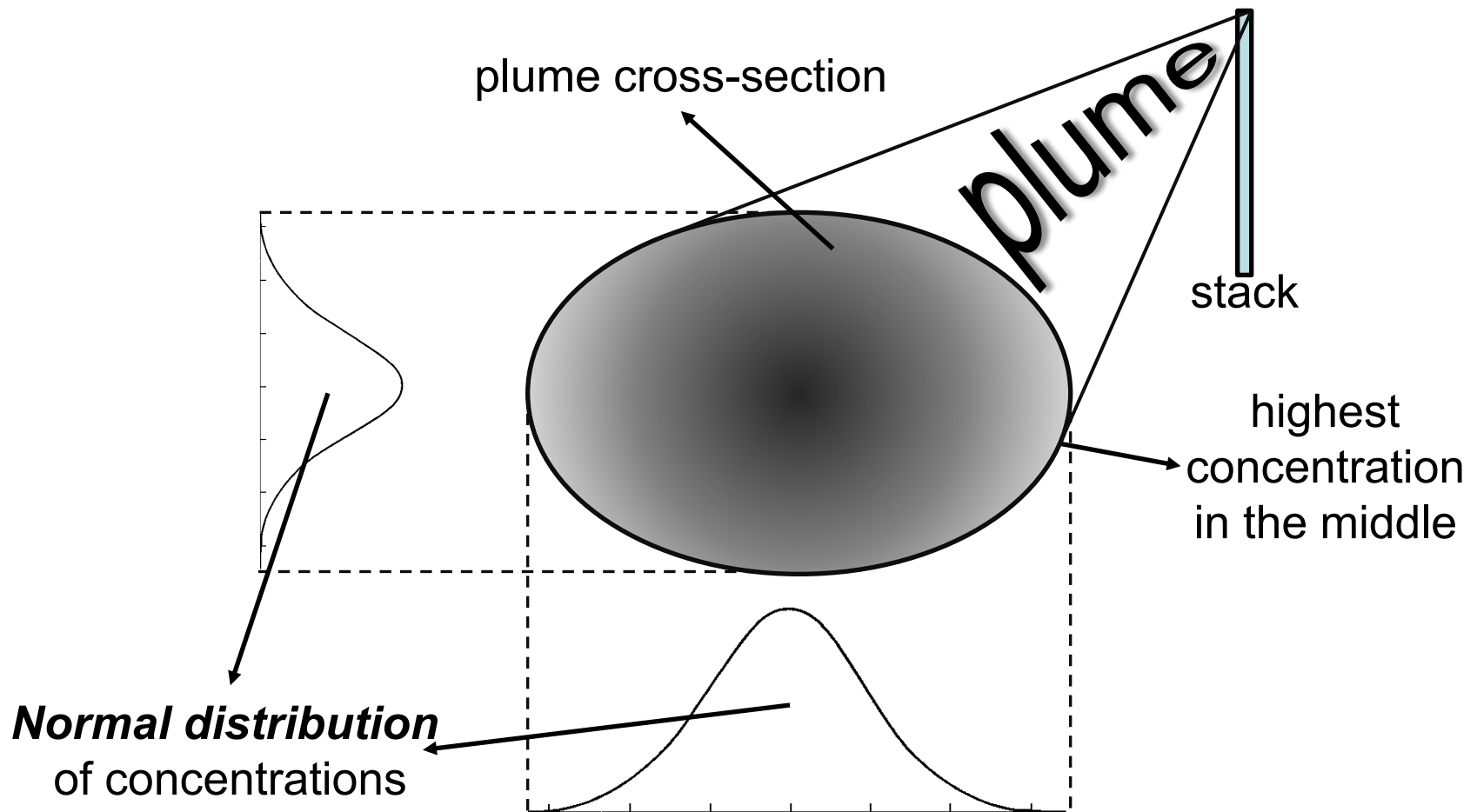
Neutral



Stable



Inside the time average plume...

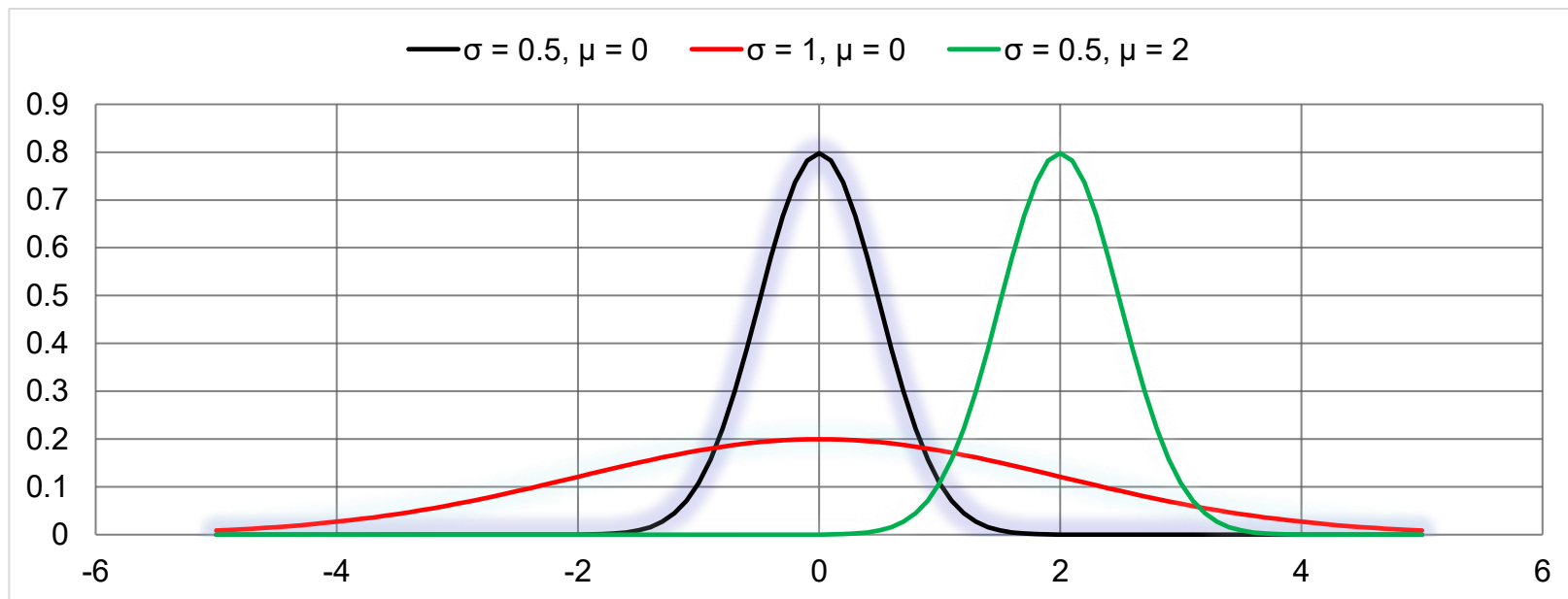


What is normal distribution

- Also called Gaussian distribution

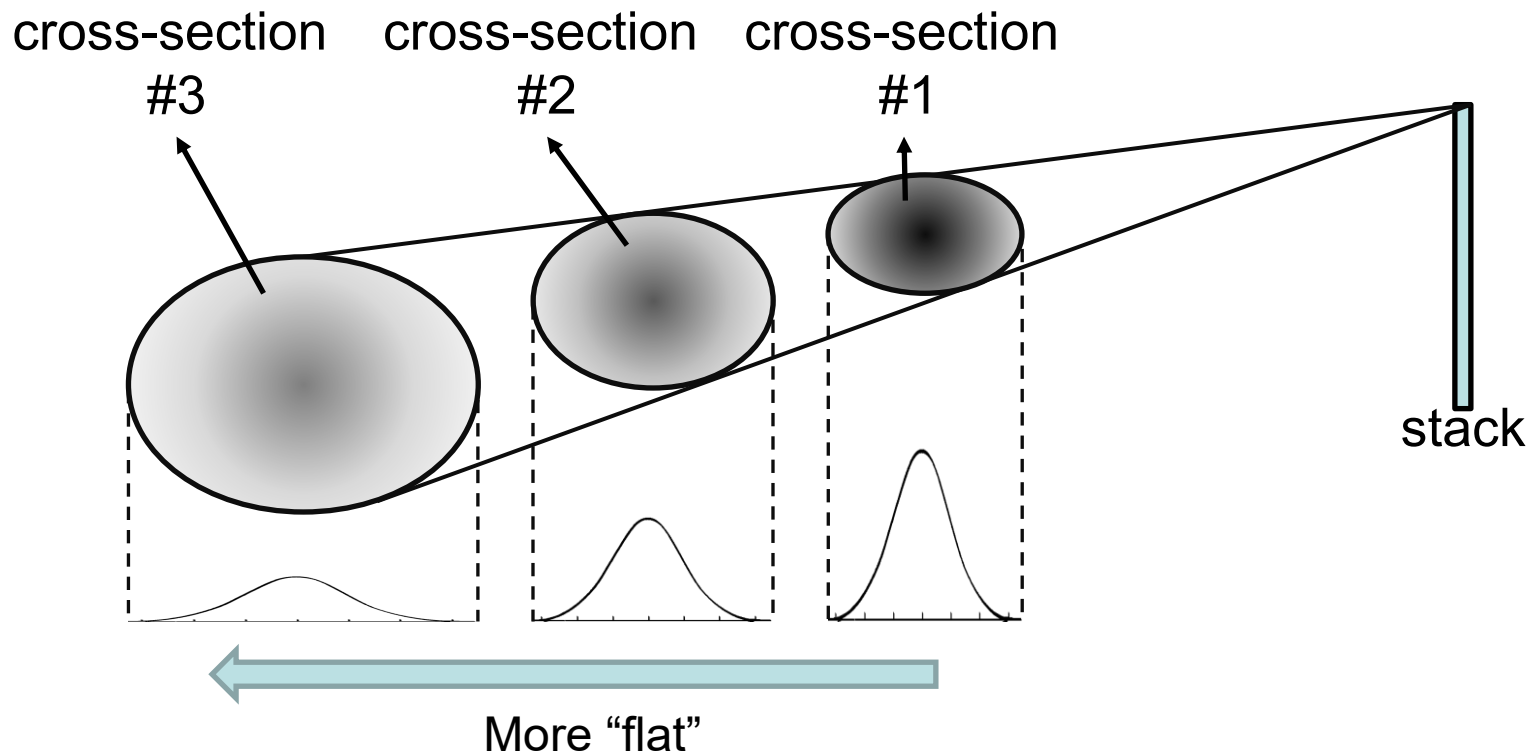
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Center of the distribution determined by μ
- “Flatness” of the distribution determined by σ

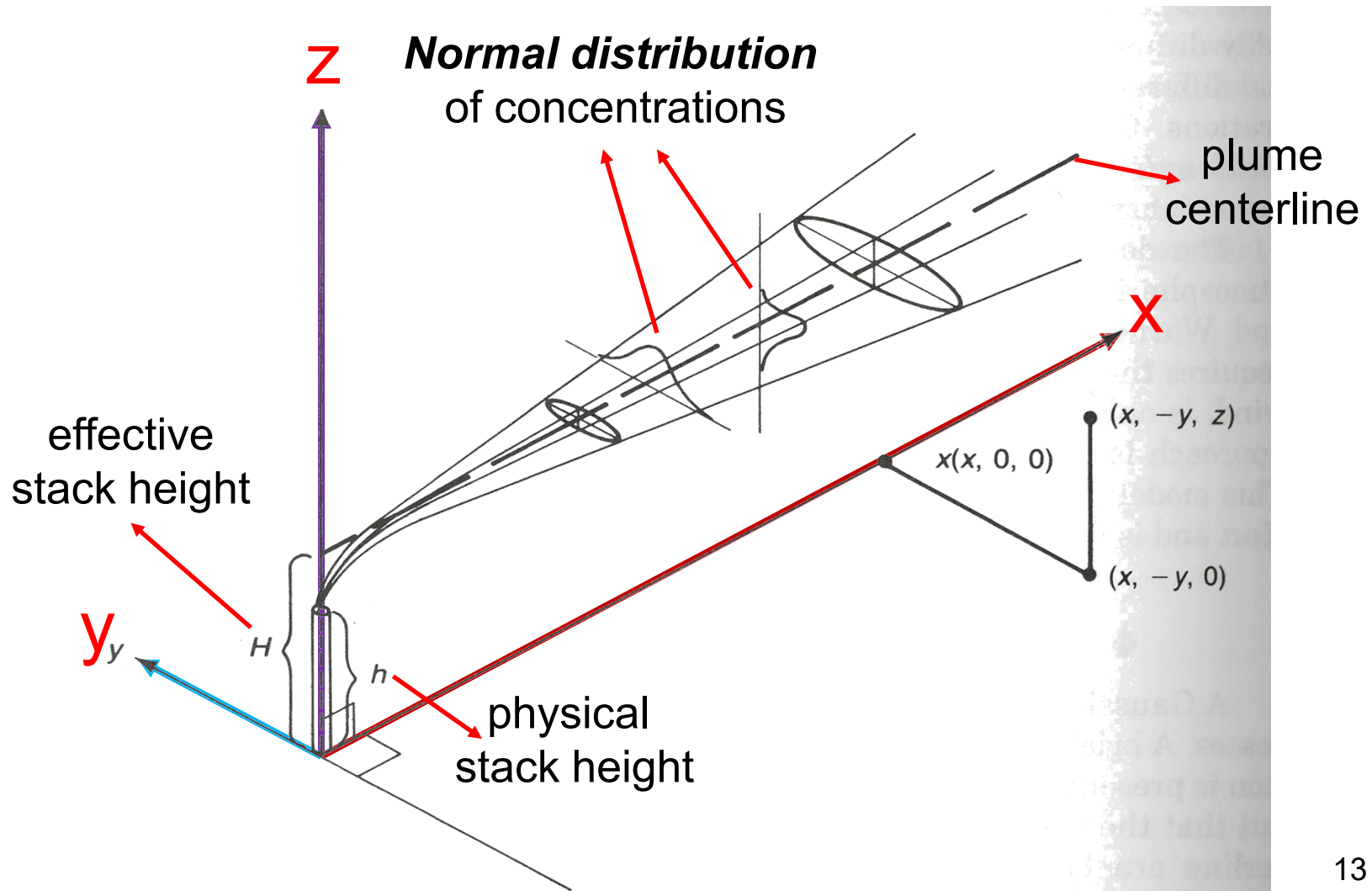


Inside the time average plume...

- The concentration distribution widens (increase σ) with increase in distance



Inside the time-average plume...



The Gaussian Plume Model

- Basic Gaussian Plume Model

Pollutant concentration at downwind location (x,y,z)

Pollutant emission rate

Effective height

$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) \right\}$$

Wind speed at effective height H

Gaussian distribution parameter, determine "flatness" at y (σ_y) and z (σ_z) direction

The Gaussian Plume Model

- Take account into ground reflection

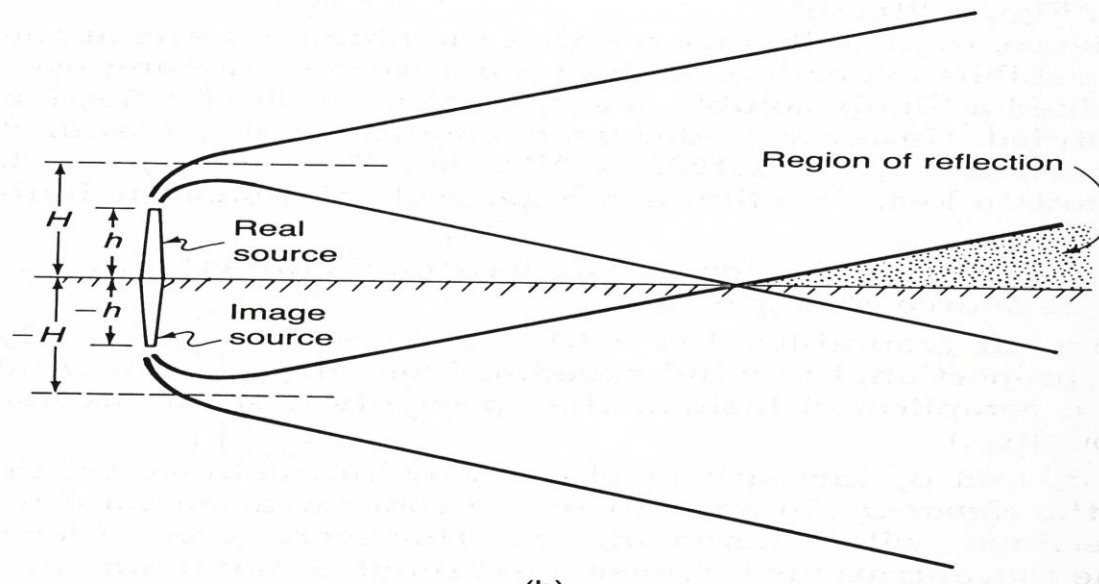


Figure 20.5 (Cooper and Alley, 2002)

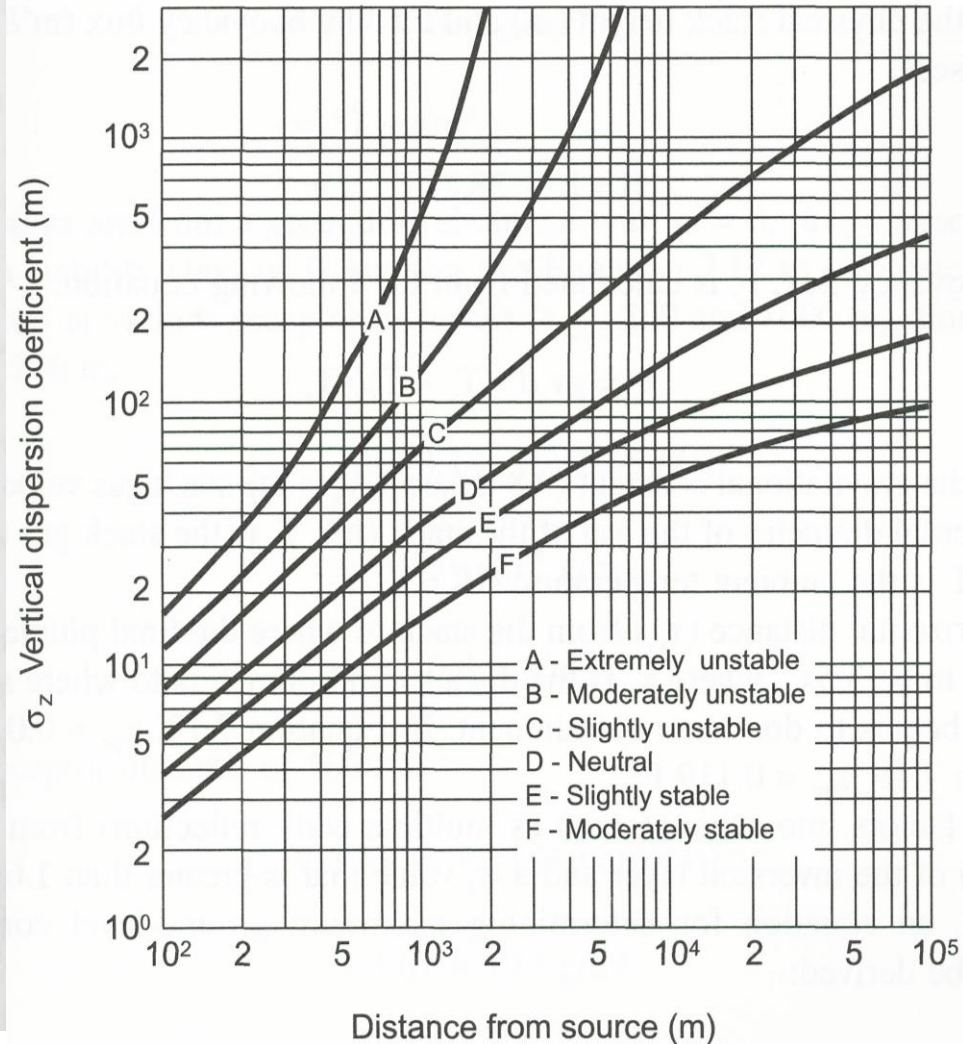
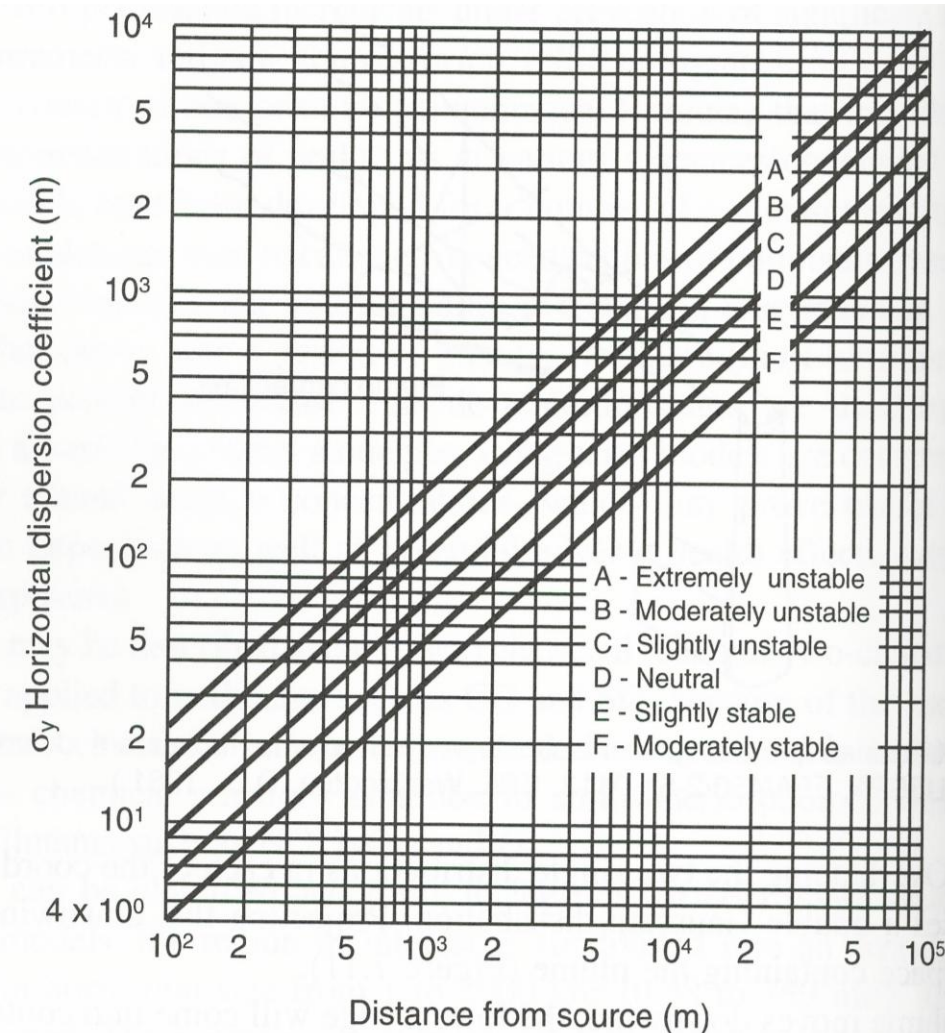
$$C(x,y,z) = \frac{S}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left\{ \exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right) \right\}$$

*This one is the **most used** form of Gaussian plume model*

How to find σ_y and σ_z

- Two methods to find σ_y and σ_z
 - Look them up in two figures
 - Calculate them using fitted equations
 - There are many different fitted equations available
 - Based on stability class and distance

How to find σ_y and σ_z (*method 1*)



(Godish, 2003)

How to find σ_y and σ_z (*method 2*)

Table 20.2 Values of Curve-Fit Constants for Calculating Dispersion Coefficients as a Function of Downwind Distance and Atmospheric Stability

Stability	<i>a</i>	<i>b</i>	<i>x</i> < 1 km			<i>x</i> > 1 km		
			<i>c</i>	<i>d</i>	<i>f</i>	<i>c</i>	<i>d</i>	<i>f</i>
A	213	0.894	440.8	1.941	9.27	459.7	2.094	-9.6
B	156	0.894	106.6	1.149	3.3	108.2	1.098	2.0
C	104	0.894	61.0	0.911	0	61.0	0.911	0
D	68	0.894	33.2	0.725	-1.7	44.5	0.516	-13.0
E	50.5	0.894	22.8	0.678	-1.3	55.4	0.305	-34.0
F	34	0.894	14.35	0.740	-0.35	62.6	0.180	-48.6

Adapted from Martin, 1976.

$$\sigma_y(x) = ax^b$$

$$\sigma_z(x) = cx^d + f$$

$$\sigma [=] \text{ m}, x [=] \text{ km}$$

Assumptions of Gaussian Plume Model

- Steady-state conditions
 - Constant wind direction, speed etc..
 - No spatial variation of wind field...
 - No background concentrations
- No terrain effect.
 - More assumptions see Abdel-Rahman, 2008 (p32)
- Mainly applied in short range ($< 50\text{km}$), flat terrain, simple meteorological conditions

Max concentration

- Maximum concentration is of interest many times
- Always occur at directly downwind ($y = 0$)

$$C_{max} = \frac{Q}{\pi u \sigma_y \sigma_z} \exp\left(-\frac{H^2}{2\sigma_z^2}\right)$$

$$\left(\frac{Cu}{Q}\right)_{max} = e^{a+b\ln H+c(\ln H)^2+d(\ln H)^3} \quad \text{Fitted equation}$$

$$\sigma_z = \frac{H}{\sqrt{2}} \text{ when neutral}$$

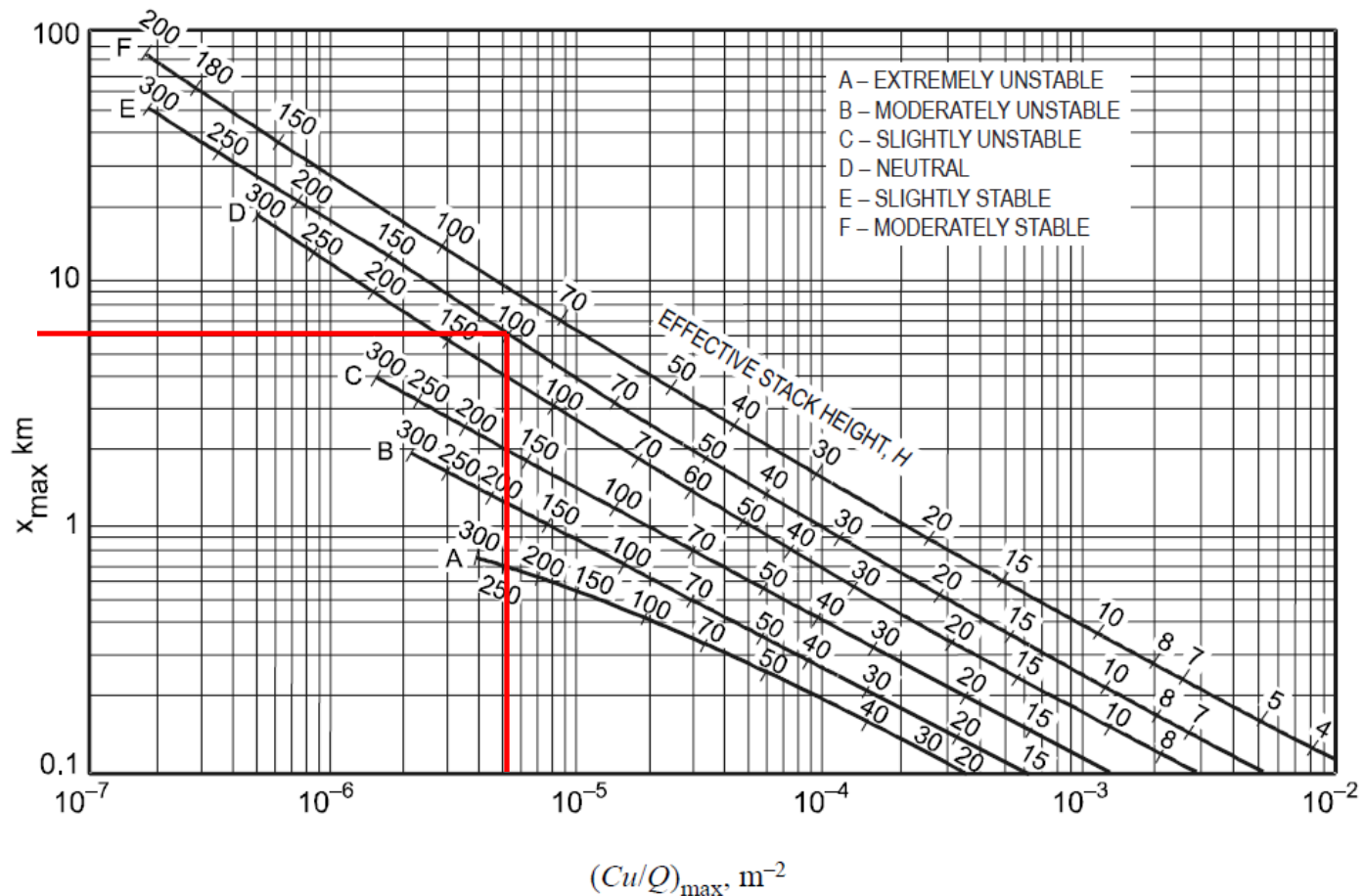
Max concentration

Stability	Constants			
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
A	−1.0563	−2.7153	0.1261	0
B	−1.8060	−2.1912	0.0389	0
C	−1.9748	−1.9980	0	0
D	−2.5302	−1.5610	−0.0934	0
E	−1.4496	−2.5910	0.2181	−0.0343
F	−1.0488	−3.2252	0.4977	−0.0765

$$\left(\frac{Cu}{Q}\right)_{max} = e^{a+b\ln H+c(\ln H)^2+d(\ln H)^3}$$

Q and u often known, C can be quickly estimated

Max concentration



Cu/Q vs X when max conc occur
(Turner, 1970)

Max conc example

Emission rate 110 g/s from a stack with effective height 100 m. Wind speed is 5 m/s. Stability class D. What is C_{max}?

- Use Cu/Q vs X figure: X_{max} is 3 km, (Cu/Q)_{max} is $8.1 \times 10^{-6} \text{ m}^{-2}$
 $C_{\max} = 8.1 \times 10^{-6} \times Q / u = 8.1 \times 10^{-6} \times 110 \div 5 = 178 \text{ } \mu\text{g/m}^3$
- Use fitted (Cu/Q)_{max} equation: a = -2.5302; b = -1.561; c = -0.0934

$$C_{\max} = \frac{110(10)^6}{5} \exp \left[-2.5302 - 1.561 \ln 100 - 0.0934 (\ln 100)^2 \right]$$
$$= 183 \text{ } \mu\text{g/m}^3$$

$$\left(\frac{Cu}{Q} \right)_{\max} = e^{a + b \ln H + c (\ln H)^2 + d (\ln H)^3}$$